SQL Fundamentals

Exercise 3: SQL CASE statements

1 Classify each product by price:

```
Select Product_name, Price
Case When Price > 1000 then 'Expensive'
     When Price between 100 and 1000 then 'Mid range'
     When Price < 100 then 'Budget'
     END AS Price_category
From Products;
```

Products_name	Price	Price_category
Laptop	1200.00	Expensive
Phone	800.00	Mid range
Keyboard	45.00	Budget
Monitor	300.00	Mid range
Mouse	25.00	Budget.

2 Label each order

```
Select Customer_name, amount
Case When order ≥ 1000 then 'High value'
     When Medium Value for order
```

2 Label each order

```
Select Customer_name, amount, order_id
    Case When order ≥ 1000 then 'High value'
         When order between 500 and 999.90 then 'Medium value'
         When low value Order < 500 then 'low value'
    END AS order_value_category
From orders;
```

Customer_name	Amount	Order_value_Category
Alice	150.00	Low Value
Bob	560.00	Medium Value
Charlie	999,99	Medium Value
Diana	45.50	low value
Ethan	1200.00	High Value

3 Categorize employee position

```
Select emp_name, Salary, department
    Case
        When department = 'IT' AND Salary > 80000 then 'Senior IT'
        When department = HR AND Salary > 55000 then 'Experienced HR'
        Else 'Staff'
    END AS Position_level
From employees';
```

emp_name	Department	Salary	Position level
John	IT	85 000	Senior IT
Sara	HR	60 000	Experienced HR
Mark	IT	75 000	Staff
Lucy	Finance	95 000	Staff
Tom	HR	55 000	Staff

4 Assign a ~~letter~~ grade

```
Select Score, grade
    Case When Score >= 90 then 'A'
         When Score between 80 and 89 then 'B'
         When Score between 70 and 79 then 'C'
         When Score between 60 and 69 'D'
         ~~Else~~ When Scae < 60 then 'F'
    END as grade
```

From Student;

Student_name	Score	grade
Anna	92	A
Ben	71	C
Cara	56	F
David	83	B
Ella	68	D

5 Label delivery performance:

```
Select delivery_id, delivery_time_minutes
    Case When time ≤ 30mins then 'fast'
         When time between 31-60mins then 'On Time'
         When time ≤ 60 mins then 'late'
    END AS performance
    From deliveries;
```

Delivery_id	Delivery_time_minutes	Performance
1	45	On time
2	60	Late
3	30	fast
4	65	late
5	100	late

6 Convert priority to labels

```
Select Issue_type, priority
    Case When Priority = 3 then High
         When Priority = 2 then Medium
         When Priority = 1 then Low
    END AS Priority_label
From tickets;
```

Issue_type	Priority	Priority_label
Login issue	1	Low
Server down	3	High
Slow system	2	Medium
Email error	2	Medium
Password reset	1	Low

7 Calculate attendance % and classify:

```
Select days_present, Student_id
  Case When Attendance >= 90% then 'Excellent'
       When Attendance between 75-89% then 'Good'
       When Attendance < 75% then 'Needs improvement'
  END AS 'Attendance_percentage', Attendance_status
from attendance;
```

Student_id	Attendance_percentage	Attendance status
1	90%	Excellent
2	60%	Needs improvement
3	96%	Excellent
4	50%	Needs improvement
5	100%	Excellent

8 Label stock status:

```
Select Product_id, Stock_qty
  Case When stock = 0 then 'out of stock'
       When stock between 0 and 5 then 'low stock'
       When stock >= 5 then 'In stock'
  END AS Stock_status
from products_inventory;
```

Product_id	stock_qty	stock_status
1	5	In stock
2	0	Out of stock
3	25	In stock
4	10	In stock
5	3	Low stock

9 Classify by size

```
Select Subject, enrolled_students
    Case when size >= 25 then 'Large'
         When size between 10 and 24 then 'Medium'
         When size < 10 then 'Small'
    END AS class_size_category
From classes
```

Subject	Enrolled_students	class_size_Category
Math	30	Large
English	25	Large
Science	15	Medium
Art	5	Small
History	20	Medium

10 Apply discount flag

```
Select Amount, payment_method, payment_id
  Case When payment_method = 'Cash' and amount >= 200
       then 'Eligible for Discount'
       Else Otherwise
```

END AS discount_eligibility
from payments;

Payment_id	payment_method	Amount	Discount_Eligib
1	Card	50.00	Not Eligible
2	Cash	200.00	Eligible for Disc
3	Card	150.00	Not Eligible
4	Paypal	75.00	Not Eligible
5	Cash	300.00	Eligible for Disca